

Dylan Miller

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EDUCATION

Texas A&M University, College of Engineering

Expected May 2030 | College Station, TX

- Bachelor of Science in General Engineering | Pursuing Entry to Mechanical Engineering | University Honors Program
- Terry Scholarship Recipient:** An extremely competitive, full-ride scholarship awarded to exceptional Texas students who demonstrate leadership, academic excellence, and financial need through a rigorous essay and interview process.

TECHNICAL EXPERIENCE

Ben Barber Innovation Academy | Engineering Design Student

Jan 2024 - May 2025 | Mansfield, TX

- Operated Haas CNC lathes and mills, maintaining 100% up-time across \$200,000+ of advanced manufacturing machinery.
- Successfully presented and defended a patent-pending Scent-Based Alarm Clock prototype before 5 industry professionals.
- Managed production with \$100,000+ worth of industrial CNC plasma tables, laser cutters, and professional 3D printers, manufacturing components for 8+ engineering projects on a weekly basis across 3 consecutive semesters.
- Maintained a 97% average across Introduction to Engineering Design (IED), Computer Integrated Manufacturing (CIM), and Engineering Design & Development (EDD) coursework.
- Designed and CNC-machined a wind-powered miniature air pump motor, machining functional components from metal stock to convert rotational wind energy into working pneumatic output, achieving full functional performance on the first test run.
- Modeled a two-car toy train in Autodesk Fusion 360, engineering the first car to strict pre-specified dimensions and independently designing a second car with custom holiday-themed geometry and lettering to demonstrate open-ended design capability.
- Utilized RoboCell Simulation software and Autodesk Inventor within the Project Lead the Way (PLTW — Recognized Texas A&M University Partner) curriculum to program and automate robotic cells before physical manufacturing.

FIRST Tech Challenge Robotics | Team Member of the "Fighting Pickles"

Aug 2024 - May 2025 | Mansfield, TX

- Placed 9th in the FIRST Tech Challenge, an international STEM competition involving 200,000+ students across 110+ countries.
- Calculated exact torque requirements and mechanical advantages to optimize a 4-motor drivetrain, increasing robot acceleration and field velocity parameters by 15%.
- Designed, built, and integrated an intake and custom lift arm mechanism in Fusion 360 using 3D-printed components, reducing total structural weight by 10%.
- Collaborated within a 10+ member cross-functional team to diagnose mechanical failures under tight time constraints, ensuring 100% hardware readiness before alliance match play.
- Programmed autonomous and driver-controlled robot functions in C#, improving reliability and match performance.

AP Physics | Team Project Member

Aug 2024 - May 2026 | Mansfield, TX

- Calibrated experimental setups, leading 15+ distinct sensor arrays, such as photogates, to capture high-frequency motion data.
- Conducted laboratory data analysis, applying rigorous propagation of error methods across multi-variable equations to isolate systematic mechanical friction from random environmental factors.
- Formulated, built, and tested 10+ physics projects and 20+ in-class labs, catalyzing learning from failure rather than success.

WORK EXPERIENCE

Family Automotive Towing & Recycling Business | Automotive Helper

May 2023 - Present | Dallas, TX

- Managed daily operations, towing dispatch, and material handling, developing practical knowledge of business management.
- Diagnosed and assisted with repairs on customer and company vehicles, building 3+ years of hands-on mechanical aptitude.

Walnut Creek Country Club | Head Lifeguard

May 2025 - Aug 2026 | Mansfield, TX

- Promoted to Head Lifeguard by demonstrating consistent leadership and performance in a high-responsibility environment.
- Interviewed, hired, and mentored 30+ lifeguards while ensuring compliance with safety and city regulations.
- Fabricated the capability to analytically and visually detect flaws in certain systems such as pool safety regulations.

LEADERSHIP EXPERIENCE

National Honor Society | Vice President and Treasurer

May 2024 - May 2026 | Mansfield, TX

- Completed 90+ community-service hours while directing service initiatives and chapter operations for 100+ members.
- Executed a record-breaking canned food drive that achieved a 33% higher contribution rate and 10,000+ can donation.
- Conducted outreach to local and chain restaurants such as Chick-Fil-A, securing a 50% donation success rate for events.

Soccer | Varsity and Club Co-Captain

Aug 2022 - May 2026 | Mansfield/Arlington, TX

- Cultivated communication and collaborative problem-solving skills by coordinating with a wide spectrum of teammates.
- Analyzed opponents' strategies to model and set examples for teammates in extreme environments.

PROFESSIONAL DEVELOPMENT

Certifications: Autodesk Fusion 360 Associate, America's Cutting Edge (ACE) CNC, OSHA 10-Hour General Industry, StarGuard Elite Adult First Aid/CPR/AED.

Competencies: Project Management, Mechanical Design, Engineering Problem Solving, Design Iteration, Technical Communication, Leadership, Collaboration, Public Speaking, Attention to Detail, Time Management, Adaptability, Customer Service, Microsoft 365.